Ayush Priyadarshi

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Research Profile

Physics student developing Geant4–Python simulation tools and angle-resolved detector-response methods for gamma-ray spectroscopy and positron-emitter coincidence summing. Seeking PhD research in detector physics, radiation detection, nuclear or particle instrumentation, and scientific computing, with broader experience in embedded hardware and Linux systems.

Education

Indian Institute of Technology Roorkee Roorkee, India

Integrated Bachelor of Science–Master of Science in Physics
Nov. 2021 – Expected Apr./May 2027 Current CGPA: 4.542/10
Thesis performance: Stage I: 8/10; Stage II: 9/10
Supervisor: Prof. Anil Kumar Gourishetty

Research & Instrumentation Experience

Detector research and thesis: Department of Physics, IIT Roorkee
Supervisor: Prof. Anil Kumar Gourishetty

Annular NaI(Tl) Annihilation-Photon Correlation Factors Aug. 2026

- Corrected an earlier unpublished derivation and developed a full-interval calculation of angular-correlation factors without assuming a direction-independent peak-to-total ratio.
- Led a one-million-event, angle-resolved Geant4 analysis; fitted detector-response coefficients and validated the physics against NIST XCOM and physical/experimental checks.
- Released the Angular Factors Calculator; first and corresponding author of a paper accepted for poster presentation at DAE SNP 2026, establishing the annular calculation for subsequent extension to the soccer-ball array.

Coincidence Summing in the TIFR Soccer-Ball NaI(Tl) Array July 2026–Present

Extending the annular approach to the 32-element near-4π array, providing modelling and angular factors for Prof. Gourishetty's ongoing β^+ correction formalism.

- Implemented custom Geant4 sources from nuclear decay schemes, including emission probabilities, characteristic gamma rays, and back-to-back 511-keV annihilation photons.
- Completed 10^8-event simulation campaigns for Na-22, O-14, and Sc-44 and calculated the array's detector-specific angular factors.
- Benchmarked the detector model against published TIFR measurements and validated the correction procedure against coincidence-free monoenergetic simulations. Journal manuscript planned; confirmed second author.

Master's Thesis Research and Software Development 2025–Present

A Python-Assisted Geant4 Workflow for Automated Scintillation Detector Response Simulations

- Developed an integrated Geant4 C++, Python automation, and ROOT analysis workflow for scintillation detector-response simulations.
- Automated simulation campaigns, output validation, spectral analysis, and peak-count extraction across detector geometries and scintillator materials.
- Executed high-statistics simulations on PARAMGanga HPC and released a reproducible public workflow with reference simulations and analysis outputs.

Research & Instrumentation Experience (continued)

Radio Telescope Control and Embedded Linux Support 2023–2024

Assisting contributor, Physics and Astronomy Club, IIT Roorkee

- Assisted with a custom Raspberry Pi image and hardware-control integration; modified boot configuration for reliable startup under lower-voltage portable power conditions.
- Enabled single-cable networking over the power/data port.
- Modified and rebuilt a Wi-Fi driver to enable compatibility with the Raspberry Pi.
- Interfaced telescope-control motors, magnetic sensors, and accelerometers with the Raspberry Pi.

Publications & Conference Contributions

A. Priyadarshi, R. Chauhan, A. Srivastava, and Anil Kumar Gourishetty. "Estimation of angular correlation factors for annihilation photons in a NaI(Tl) annular detector." DAE Symposium on Nuclear Physics, 7–11 December 2026.
Accepted — Poster Presentation; First & Corresponding Author

A. Priyadarshi, G. A. Kumar, K. Madhan, and V. Ranga. "A Python-based Tool for Automated Geant4 Simulations of Scintillation Detectors." IEEE Nuclear Science Symposium, 2026.
Accepted — Poster Presentation; First & Corresponding Author

Awards & Grants

- **Paul Phelps Continuing Education Grant**, IEEE NSS/MIC/RTSD; Sept. 2026.
- **Category Winner**, Solar Decathlon India Net-Zero Building Competition; June 2026. Rs. 1,00,000 cash prize.
- **First Prize, Alumni Track**, Syntax Error Hackathon, SDS Labs, IIT Roorkee; March 2022. Built a Raspberry Pi-based WiFi IP Mouse; Rs. 10,000 cash prize.

Research Software

Angular Factors Calculator | Developer and Maintainer GitHub
Python package and CLI for annular angular-correlation factors from angle-resolved ROOT events or supplied mass coefficients; numerical cross-checks, structured outputs, tests, and continuous integration.

Geant4 Detector Simulation Pipeline | Developer and Maintainer GitHub
Configurable simulation and analysis workflow for scintillation detector studies, with reproducible spectra, count tables, and public Cs-137 reference demonstrations.

Technical & Instrumentation Skills

Programming: Python, C++, Bash, Fortran, SQL
Scientific computing: Geant4, ROOT, Monte Carlo, detector-response modelling
HPC: PARAMGanga (IIT Roorkee's supercomputer); batch workflows and high-statistics simulations
Systems & tools: Linux administration, Docker, networking, Git, Qt, MySQL
Embedded hardware: Raspberry Pi, Arduino, ESP8266, FPGA, relay circuits
Radiation detection: NaI(Tl), $LaBr_3$, gamma spectroscopy, radioactive sources, signal acquisition
Nuclear electronics: NIM/VME modules, pulse processing, counting systems, oscilloscopes

Selected Personal Projects

Raspberry Pi Home Server and Network Gateway 2023–Present

Designed and maintain a Linux/Docker server for self-hosted services, file sharing, home automation, DNS filtering, reverse proxy services, and network gateway functionality.

IoT Smart Switch with Alexa Integration 2020

Built an ESP8266-based Wi-Fi smart switch with relay-controlled switching for household lights and fans through Alexa voice commands.